

SALES INTELLIGENCE

End-to-End Analytics Engineering Project | Business Context Document

Total Revenue	Total Orders	Gross Margin	Unique Customers	Active Products	Countries
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Business Definition

Overview

A retail business selling bikes, accessories, and clothing across six international markets (United States, Australia, United Kingdom, Germany, France, and Canada) had no unified view of its sales performance. Raw transactional data was fragmented across two source systems — a CRM and an ERP — stored as CSV files with quality issues including duplicates, mismatched keys, missing demographic fields, and inconsistent geographic formats.

Core Business Problem

Without a clean, integrated data pipeline, the business was unable to:

- Track revenue trends or identify demand seasonality reliably across its trading history.
- Understand which products, regions, or customer segments generated the most value.
- Diagnose margin inefficiency or measure concentration risk across its product portfolio.
- Make budget or geographic expansion decisions backed by a trustworthy, single source of truth.

Analytical Scope

This project spans multiple transactions over several fiscal years. The scope covers product performance across three categories (Bikes, Accessories, Clothing) and four product lines (Road, Mountain, Touring, Sport), customer intelligence across a large base of unique buyers, and geographic performance across six countries.

Key Business Questions Answered

The following three strategic questions drove the analytical design of this project. Each was answered through the data pipeline and exploratory analysis.

#	Strategic Question	Analytical Focus
1	Where is revenue growth coming from — and what structural risks threaten it?	Examining revenue trends across four years and six geographies to understand growth drivers, seasonal patterns, and the risks posed by geographic concentration, category dependency, and discontinued product reliance.
2	Which products and categories generate the most value — and where is margin being lost?	Assessing gross margin performance across product categories, lines, and individual SKUs to surface the paradox between revenue concentration (Bikes) and margin efficiency (Accessories), and to identify high-ROI optimisation levers.
3	Who are the most valuable customer segments — and how should acquisition and retention be prioritised?	Segmenting customers by age group, gender, geography, and purchase behaviour to identify high-value cohorts, understand lifetime spend patterns, and guide targeted retention and acquisition investment.

Data Sources & Tables Used

The project ingests data from two source systems (CRM and ERP). The tables below represent the primary entities used throughout the pipeline.

Table / View Name	Description	Layer
crm_cust_info	Customer master data — IDs, names, demographics (CRM source)	Bronze / Silver
crm_prd_info	Product catalogue — names, categories, costs, lifecycle dates (CRM source)	Bronze / Silver
crm_sales_details	Transactional sales records — order IDs, quantities, prices (CRM source)	Bronze / Silver
erp_cust_az12	Customer demographic supplement — gender, birth dates (ERP source)	Bronze / Silver

Table / View Name	Description	Layer
erp_loc_a101	Geographic lookup — customer country codes and names (ERP source)	Bronze / Silver
erp_px_cat_g1v2	Product category and subcategory mapping (ERP source)	Bronze / Silver
dim_customers	Cleaned, enriched customer dimension — resolved gender, geography, age group	Gold (View)
dim_products	Deduplicated product dimension — active flag, category, cost, line	Gold (View)
fact_sales	Integrated sales fact table — sales amount, quantity, order dates, keys	Gold (View)
report_sales_details	Flat denormalised table for consumption — all dimensions joined	Gold (View)
report_customers	Aggregated customer-level metrics — lifetime value, order count, lifespan	Gold (View)
report_product_performance	Product-level performance — total customers, orders, revenue by SKU	Gold (View)